

Exercise 1

Let X be a topological space, and suppose $Z \subseteq Y \subseteq X$. Let Y have the relative topology as a subset of X . Prove that the topology on Z as a subset of Y is the same as its topology as a subset of X .

Proof: Suppose $U \subseteq Z$ is open in the relative topology from $Z \subseteq Y$. So $U = V \cap Z$ for some V open in Y . Moreover, V being open in Y means $V = W \cap Y$ for some W open in X . Then

$$U = W \cap X.$$

Thus U is open in the relative topology on X .

Conversely, suppose $U \subseteq Z$ is open in the relative topology from $Z \subseteq X$. This means $U = W \cap Z$ for some W open in X . Then

$$U = (W \cap Y) \cap Z$$

where $W \cap Y$ is open in the relative topology on Y . Thus U is open in the relative topology on Y . ■

Exercise 2

The usual topology on $\mathbb{R}$ has base open intervals (a, b) . Let the **lower limit topology** have base intervals of the form $[a, b)$. Prove that open sets in the usual topology are also open in the lower limit topology.

Proof: Let $U \subseteq \mathbb{R}$ be open in the usual topology. Let $u \in U$ be arbitrary.

Then $u \in (a, b) \subseteq U$ for some basic open (a, b) . Thus $u \in [u, b) \subseteq U$, so U is open in the lower limit topology. ■

The converse is not true. For example, let $U = [0, 1)$ be an open set in the lower limit topology. If U were open in the usual topology, then

$$0 \in (a, b) \subseteq U$$

for some basic open (a, b) where $a < 0 < b$.

In this case, $a/2 \in (a, b)$ yet $a/2 \notin [0, 1)$. This is a contradiction, so U cannot be open in the usual topology.

Exercise 3

Prove that X is a Hausdorff topological space if and only if

$$\Delta = \{(x, x) : x \in X\}$$

is closed in $X \times X$.

Proof: Suppose X is Hausdorff. The notetaker has not yet proven the forward direction, but she swears it's true.

Conversely, suppose Δ is closed. This means $(X \times X) \setminus \Delta$ is open.

Let $x, y \in X$ be distinct points. Then $(x, y) \notin \Delta$, so (x, y) is a point in $(X \times X) \setminus \Delta$, an open set. Thus

$$(x, y) \in U \times V \subseteq (X \times X) \setminus \Delta$$

for some basic open set $U \times V$.

Assume $z \in U \cap V$. Then (z, z) is a point in $(U \times V) \cap \Delta$, a contradiction. So $x \in U$, which is open; $y \in V$, which is also open; and $U \cap V = \emptyset$. Thus X is Hausdorff. ■

Exercise 4

Let X be a topological space. Prove that the diagonal map

$$f : X \rightarrow X \times X, \quad x \mapsto (x, x)$$

is continuous.

Proof: Let $W \subseteq W \times W$ be an arbitrary open set, and let $x \in f^{-1}(W)$ be an arbitrary point in the preimage. Hence $f(x) \in W$. Since W is open,

$$f(x) \in U \times V \subseteq W$$

for some basic open $U \times V$.

Expanding $f(x)$, we have $(x, x) \in U \times V$. So $x \in U$ and $x \in V$, meaning $x \in U \cap V$.

Let $y \in U \cap V$. Then $(y, y) \in U \times V$, so

$$(y, y) \in W \implies f(y) \in W \implies y \in f^{-1}(W).$$

Therefore,

$$x \in U \cap V \subseteq f^{-1}(W). \quad \blacksquare$$

Exercise 5

Prove if X is a compact topological space and $f : X \rightarrow \mathbb{R}$ is continuous, then $f(X)$ is bounded.

Proof: Suppose $x \in X$. Let n be an absolute bound for $f(x)$. That is,

$$n = [|f(x)|].$$

Then $f(x) \in (-n, n)$, so $x \in f^{-1}(-n, n)$. Since f is continuous, the preimage $f^{-1}(-n, n)$ is open for any n . Hence

$$X = \bigcup_{n=1}^{\infty} f^{-1}(-n, n)$$

is an open cover of X . Since X is compact, it has a finite subcover

$$X = f^{-1}(-n_1, n_1) \cup \dots \cup f^{-1}(-n_k, n_k).$$

Let $N = \max\{n_1, \dots, n_k\}$. Then $X = f^{-1}[-N, N]$. Therefore,

$$f(X) \subseteq [-N, N]$$

meaning f is bounded. ■